REVIEW 1 & 2 · SETUP, SRS & WORKING APPLICATION EVALUATION

TQM Pharm Enterprise

Clinical & Retail Pharmacy Operating System with Total Quality Management Architecture
Quality Goal: Q01 Improve Reliability & Zero-Defects

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EVALUATION SCOPE:

Amrapali University

B.Tech CSE Project Reviews 1 & 2
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RUBRIC: PROBLEM STATEMENT & BASELINE UNDERSTANDING

Problem Statement & Baseline Clinical Understanding

1. Medication Dispensing Errors

Clinical Baseline:

- Over 1.3 million preventable patient injuries occur annually due to adverse drug events (ADEs).
- Manual interpretation of doctor prescriptions causes dosage miscalculations and transcription slips.
- Lack of automated allergy cross-checks risks fatal Penicillin anaphylaxis.
- Baseline Solution: Automated Clinical Decision Support System (CDSS) acting as an immutable quality shield.

2. Cold-Chain & Expiry Spoilage

Supply Chain Baseline:

- Vaccines, insulin, and biologics degrade silently outside the mandatory 2.0°C–8.0°C envelope.
- Standard retail FIFO picking allows older batches to sit unnoticed while newer stock is sold.
- Substantial annual capital lost in expired, unsellable inventory.
- Baseline Solution: Algorithmic FEFO (First-Expiry, First-Out) picking + 24/7 continuous IoT temperature telemetry.

3. Regulatory & Till Shrinkage

Operational Baseline:

- Inadequate tracking of Schedule X / NDPS controlled sedatives invites severe legal penalties and license cancellation.
- Cash drawer variances, unmonitored mid-shift payouts, and unverified digital receipts.
- Failed surprise drug inspector audits due to paper registers.
- Baseline Solution: Mandatory dual-PIN narcotic register + Daily Z-Report mathematical cash reconciliation.

RUBRIC: QUALITY OBJECTIVE

Quality Objectives Aligned with TQM Principles

Objective 1: Zero Medication Dispensing Defects (Patient Focus)

Poka-Yoke error-proofing: Automated Clinical Decision Support System (CDSS) intercepts allergen cross-reactions (Penicillin, NSAIDs) and toxic drug interactions before an invoice can be created.

Objective 2: 100% Cold-Chain Storage & Transport Compliance

Biological Integrity: Guarantee all thermo-sensitive pharmaceuticals remain in 2.0°C–8.0°C via 30s background IoT telemetry streaming and automated thermal spike alarms.

Objective 3: Zero Expiry Dead-Stock Losses (FEFO Scheduling)

Process Quality: Eliminate inventory wastage by algorithmically enforcing First-Expiry, First-Out (FEFO) batch prioritization; hard-lock dispensing of expired SKUs.

Objective 4: Zero Cash Register Discrepancy & Till Balancing

Fact-Based Management: Enforce strict shift-end financial governance through the Daily Z-Report engine, mathematically balancing opening float, sales, payouts, and counted cash.

Objective 5: 100% Regulatory Audit Readiness (ISO 9001 & CDSCO)

Continuous Assurance: Maintain an immutable 21 CFR Part 11 SQLite audit trail, dual-PIN Schedule X narcotic register, and 1-click CSV export ready for State Drug Inspectors.

RUBRIC: SCOPE AND USER ROLES

Scope and User Roles (Role-Based Access Control)

👑 Store Admin

Terminal: Console HQ

- Manage staff accounts & passwords
- Monitor live online terminal presence
- Review daily Z-Report reconciliations
- Execute SQLite snapshot backups
- Inspect immutable system audit stream

🩺 Clinical Pharmacist

Terminal: Clinical Bay

- Review incoming e-Prescriptions queue
- CDSS allergy shield cross-checks
- Drug-drug interaction risk analysis
- Schedule X narcotic PIN sign-off
- Manage chronic patient refill cycles

💳 Cashier (POS)

Terminal: Billing Counter

- 1-Click verified eRx import to cart
- Auto FEFO nearest-expiry batch pick
- Split payments (Cash, UPI QR, Card)
- Calculate cash change return
- Print 80mm GST thermal receipt

📦 Inventory Manager

Terminal: Supply Hub

- Multi-batch FEFO stock ledger
- Record Goods Receipt Notes (GRN)
- 24/7 Cold-chain IoT fridge monitor
- Manage supplier directory & POs
- Track low-stock alerts (≤20 units)

🛵 Delivery Agent

Terminal: Logistics Bay

- Active dispatch Kanban route board
- Verify cold-bag insulated packaging
- 4-Digit customer OTP verification
- Record electronic Proof-of-Delivery
- Handover timestamp audit log

RUBRIC: MODULAR FOLDER ARCHITECTURE

Modular Folder Architecture & 6-Tier System Blueprint

📁 ENTERPRISE MODULAR REPOSITORY LAYOUT

```
Pharmacy_Management_System/
├── core/
│   ├── bridge.py       # RPC Bridge (38 typed methods)
│   ├── db.py           # SQLite persistence (14 tables)
│   ├── api_client.py   # Telemetry & presence client
│   └── logger.py       # 21 CFR Part 11 audit logger
├── config/
│   └── settings.py     # Central Singleton configuration
├── ui/
│   ├── desktop/        # Edge Chromium WebView2 60 FPS
│   └── web/            # Cloud-ready FastAPI Web UI
├── apps/               # Admin & operational workstations
├── tests/              # 24 Automated pytest test cases
├── backups/            # Automated SQLite snapshots
├── main.py             # Hardware-accelerated entrypoint
└── web_app.py          # Asynchronous FastAPI web server
```

⭐ THE ONE RULE TO REMEMBER

"The screen must not talk to the database."

UI → Service (Bridge) → Model / Database

- No raw SQL queries inside billing.py or any UI template.
- All database operations are mediated by typed Python methods.
- Failure isolation: UI bugs cannot corrupt database ACID commits.
- Validation and audit have dedicated architectural homes.
- Allows layer-by-layer agile development and test-driven verification.

RUBRIC: SRS (REQUIREMENTS ENGINEERING)

Software Requirements Specification (SRS) — IEEE 830

📋 CORE FUNCTIONAL REQUIREMENTS (FR-01 TO FR-10)

- FR-01 (Auth & Telemetry): SHA-256 password hash, session UUID, 5s heartbeat.
- FR-02 (CDSS Safety Engine): eRx cross-check for patient Penicillin allergies.
- FR-03 (Schedule X Narcotics): Doctor DMC Reg + Pharmacist 4-digit PIN log.
- FR-04 (FEFO Batch Picker): Automatic sorting by earliest expiry; lock expired lot.
- FR-05 (POS Express Billing): 1-Click eRx import, cash change calc, dynamic UPI QR.
- FR-06 (Goods Receipt GRN): Inward supplier consignments with batch & cost.
- FR-07 (IoT Cold-Chain): Monitor 2-8°C sensors; trigger audible/visual excursion alert.
- FR-08 (Delivery Fleet): Kanban dispatch board with customer 4-digit OTP e-POD.
- FR-09 (Z-Report Shift Close): Mathematical cash balancing & signed Z-slip archive.
- FR-10 (Audit Logging): 21 CFR Part 11 append-only tamper-evident event stream.

⚡ NON-FUNCTIONAL REQUIREMENTS & STANDARDS

NFR-01 (Performance & Latency): UI click response <50ms; SQLite transaction commit <10ms; 60 FPS hardware acceleration.

NFR-02 (ACID Reliability): 100% ACID compliance with automatic daily snapshot backups in backups/.

NFR-03 (Security & Non-Repudiation): Role-based route shields, session timeout, and immutable audit logging.

NFR-04 (Statutory Compliance): Indian Drugs & Cosmetics Act 1940 (Schedule H1/X) and GST electronic tax calculation.

Formal SRS Artifacts: Submitted as IEEE Std 830-1998 compliant DOCX (`SRS_TQM_Pharm_Enterprise.docx`) and PDF (`SRS_TQM_Pharm_Enterprise.pdf`).

RUBRIC: MODULES AND FEATURES

Modules and Features Overview

🩺 Clinical Station & CDSS
eRx intake queue, allergy cross-matching, drug interaction warning modal, Schedule X double sign-off register.

💳 POS Express Billing Counter
1-Click approved Rx import, barcode scanner, FEFO batch selection, split payments (Cash/UPI/Card), GST receipt.

📦 Supply Chain & Inventory Hub
Multi-batch FEFO ledger, low stock alerts (≤20), Goods Receipt Notes (GRN) inwarding, supplier directory.

❄️ Cold-Chain IoT Telemetry (2-8°C)
Live circular temperature gauges for Chambers A & B, 24/7 background stream, simulated temperature spike alarm.

🛵 Delivery Fleet & Fulfilment
Active dispatch Kanban board, cold-bag insulation checklist, manual route creation, 4-digit OTP e-POD.

👑 Admin Command Center
Staff user CRUD with role assignment, active terminal presence heartbeats, Daily Z-Report till close, audit trail.

RUBRIC: CORE FEATURES/MODULES ALIGNED WITH QUALITY GOALS

Core Features/Modules Aligned with Quality Goals

Implemented Core Feature / Module	Aligned TQM Quality Goal & Principle
CDSS Allergy & Drug-Drug Shield	Zero Dispensing Defects / Poka-Yoke Error Proofing (Shingo)
Algorithmic FEFO Batch Sequencing	Zero Inventory Dead-Stock / Process Waste Elimination
24/7 IoT Cold-Chain Monitoring (2-8°C)	Zero Biological Degradation / Quality Assurance
Daily Z-Report Mathematical Balancing	Zero Till Variance / Fact-Based Management (Juran)
Statutory Schedule X Dual PIN Register	Zero Diversion / Indian Drugs & Cosmetics Act 1940 Compliance
Immutable Audit Logs (21 CFR Part 11)	Complete Traceability, Accountability & Non-Repudiation
Automated Error Telemetry (E01-E05)	Root-Cause Prioritization (Pareto 80/20) & FMEA Risk Matrix
Customer & Patient EHR Allergy Profile	Customer & Patient Focus (Crosby Zero-Defects Standard)

RUBRIC: DATABASE/SYSTEM DESIGN

Database/System Design & Relational Data Dictionary

CORE RELATIONAL ENTITIES (PART 1)

- **1. staff (users):** Credentials, assigned RBAC role, branch, active status.
- **2. medicines:** SKU, generic, batch lot, expiry, days left, stock, MRP, rack.
- **3. prescriptions:** Patient name, doctor, items JSON, CDSS status, clinical notes.
- **4. invoices (sales):** Official tax invoices, cashier, subtotal, GST split, total amount.
- **5. delivery_orders:** Dispatch route, address, rider, cold-bag check, 4-digit OTP.
- **6. patients (customers):** EHR profiles, contact phone, chronic diseases, recorded allergies.
- **7. suppliers:** Distributor directory, GSTIN, contact phone, payment terms.

COMPLIANCE & TELEMETRY ENTITIES (PART 2)

- **8. purchase_orders:** GRN inwarding, supplier, batch items JSON, total amount.
- **9. narcotic_register:** Schedule X: Doctor DMC Reg No, Pharmacist PIN, batch, qty.
- **10. temperature_logs:** 24/7 cold-chain telemetry: probe ID, temperature (2-8°C).
- **11. refill_schedules:** Chronic maintenance therapy: last dispensed, next due date.
- **12. active_sessions:** Live presence: user ID, terminal name, 5s heartbeat timestamp.
- **13. audit_logs:** 21 CFR Part 11 append-only log: user, role, action, details.
- **14. z_reports:** Daily till close: opening float, sales, payouts, live variance.

RUBRIC: DATABASE INTEGRATION

Database Integration: ACID Transactions & Migrations

1. ACID Transaction Integrity (Atomicity & Consistency)

Transactions guarantee complete success or complete rollback: saving an invoice, cutting batch stock, and writing the audit row occur together or not at all.

2. SQLite WAL (Write-Ahead Logging) & Concurrency

Enabled Write-Ahead Logging allows simultaneous non-blocking reads from multiple terminal UI threads while writes commit in <10ms.

3. Relational Foreign Keys & Check Constraints

Foreign keys enforce referential integrity between prescriptions, invoices, and patients. Check constraints prevent negative inventory.

4. Schema Migrations Without Data Loss

The migration subsystem dynamically applies ALTER TABLE schema updates without wiping student or patient test records.

5. Automated Database Snapshots & Archival

System creates timestamped snapshot backups in backups/ daily and on application exit for disaster recovery.

RUBRIC: CRUD OPERATIONS

CRUD Operations Implementation Across All Entities

Entity / Subsystem	👑 Store Admin	🩺 Pharmacist	💳 Cashier POS	📦 Inventory	🛵 Delivery Rider
Staff & RBAC Accounts	CRUD (Add/Edit Users)	R (Staff Directory)	R (Read Profile)	R (Read Profile)	R (Read Profile)
Medicines & Batch Stock	CRUD (Full Catalog)	R (Clinical View)	R, U (Dispense Deduct)	CRUD (Stock Adjust)	R (Stock Query)
Prescriptions Intake Queue	R (Audit Review)	CRUD (Validate/Reject)	R, U (Mark Dispensed)	R (Consignment)	R (Order Lines)
Invoices & POS Billing	R (Sales History)	R (Rx Dispensing)	CR (Generate & Print)	R (Inward Bills)	R (Delivery Note)
Delivery Orders & Logistics	R, D (Fleet Admin)	R (Dispense Dispatch)	C (Dispatch Order)	R (Cold Checklist)	RU (OTP Validate)
Narcotic Register (Sched X)	R (Statutory Export)	CR (Double Sign PIN)	R (Read Only)	R (Inward Only)	No Access (Blocked)
Daily Z-Report Close	R, U (Audit Sign-Off)	No Access (Blocked)	CR (Shift Cash Close)	No Access (Blocked)	No Access (Blocked)

RUBRIC: INITIAL UI/WORKFLOW

Initial UI & Workstation Interface Design

1. Clean Dual Theming (Default: Light / White Mode)

Engineered for hospital and dispensary lighting: default white theme with crisp slate borders and an instant 1-click Dark Mode toggle.

2. Workstation Persona Login Portal Grid

Interactive visual tiles for all 5 roles pre-filling credentials instantly and assigning sessions to physical terminal counters.

3. Zero-Lag 60 FPS Edge Chromium WebView2 Rendering

Native desktop hardware acceleration eliminates UI frame freezing; background asynchronous threads handle I/O.

4. Responsive Dual Deployment Parity

Desktop application (Edge WebView2) and Web application (FastAPI) share 100% UI feature parity on tablets and laptops.

RUBRIC: MAIN SYSTEM WORKFLOW

Main System Workflow: End-to-End Pipeline

STEP 1

Stage 1: eRx Intake

- Doctor prescription uploaded
- Patient EHR allergies scanned
- Enters pending review queue

STEP 2

Stage 2: Clinical Gate

- Pharmacist runs CDSS scan
- Allergy & interaction checks
- Schedule X PIN authenticated
- Status updated to 'Approved'

STEP 3

Stage 3: POS Express

- 1-Click import to Cashier cart
- Auto FEFO batch picking
- Dynamic UPI QR / Cash change
- Thermal 80mm GST receipt printed

STEP 4

Stage 4: Cold Fleet

- Cold-bag checklist verified
- Dispatched on Kanban route
- Customer gives 4-digit OTP
- Marked 'Delivered' (e-POD)

STEP 5

Stage 5: Z-Report Close

- Cash drawer counted
- Variance calculated (₹0.00)
- Shift signed & closed in SQLite
- CSV exported for accounting

🔄 ARCHITECTURAL PIPELINE SUMMARY: HOW A CLICK TRAVELS

1. Screen takes user click → 2. Service checks role authorization → 3. CDSS verifies patient allergy EHR → 4. Validators assert stock ceiling & expiry → 5. SQLite executes atomic commit (bill + cut stock) → 6. Audit log row written → 7. Dashboard KPI tickers update.

TQM ROOT-CAUSE ANALYTICS

TQM Quality Analytics: Pareto (80/20) & Fishbone

📈 PARETO ROOT-CAUSE ANALYSIS (80/20 RULE)

1. Look-Alike / Sound-Alike Drugs (45%): Addressed by CDSS generic molecule search and allergy confirmation.

2. Qty > Available Stock Attempts (30%): Addressed by pre-save inventory balance assertion locks.

3. Near-Expiry Drug Dispensing (15%): Addressed by algorithmic FEFO nearest-expiry prioritization.

4. Insulated Cold-Bag Packaging (7%): Addressed by mandatory delivery pre-dispatch checklist.

Key TQM Takeaway: Eliminating the top 2 error modes removes 75% of all operational dispensary defects.

🐟 FISHBONE (ISHIKAWA) ROOT-CAUSE BREAKDOWN

People (Staff): Rushed cashiers during counter peaks; new dispensers misinterpreting doctor handwriting.

Process (Workflow): Lack of second-check clinical verification step before POS billing.

Software (System): Legacy systems allowing expired stock cart addition and lacking allergy alerts.

Data (Integrity): Missing batch numbers, incorrect physical expiry dates, unrecorded patient allergies.

Machine (Hardware): Slow workstation hardware and missed clicks on crowded UI screens.

TQM QUALITY GOVERNANCE

TQM Quality Governance: FMEA Risk & Deming PDCA

⚠️ FMEA TABLE (RPN = S x O x D)

Sell Expired Drug: S=10, O=4, D=3 → RPN 120

Action: Hard block prevents cart checkout.

Dispense Allergen Drug: S=10, O=3, D=4 → RPN 120

Action: CDSS allergy warning & override block.

Qty > Stock Oversell: S=8, O=5, D=3 → RPN 120

Action: Pre-commit inventory balance assert.

Duplicate Barcode Scan: S=6, O=2, D=2 → RPN 24

Action: Unique constraint on SKU barcode.

🔁 DEMING PDCA QUALITY CYCLE

PLAN: Identify excessive oversell and near-expiry mistakes during peak rush.

DO: Deploy automated FEFO batch picker and pre-save stock balance locks.

CHECK: Monitor monthly counts of error codes E01 and E03 in error_logs.

ACT: Standardize FEFO logic across all terminals as mandatory baseline policy.

Log Feedback Loop: Blocked actions write to error_logs → tqm_report counts frequency → Occurrence score dynamically updates in FMEA table.

RUBRIC: WORKING APPLICATION

Working Application Setup, Verification & Personas

🚀 APPLICATION LAUNCH & VERIFICATION

Automated Tests (Quality Proof): `pytest`

→ 24/24 unit & integration tests passing in ~7.8s.

Desktop Workstation App: `python main.py`

→ Boots hardware-accelerated Edge Chromium WebView2 window.

Cloud Web Portal: `python web_app.py`

→ Starts FastAPI server at http://127.0.0.1:8000 with Swagger docs.

Terminal Binding: Assign session to Front Counter A, Clinical Kiosk, or Dock.

🔑 WORKSTATION PERSONAS & CREDENTIALS

👑 **Store Admin: `admin` / `admin123`**

Scope: RBAC, live presence, Daily Z-Report balancing

🩺 **Pharmacist: `pharmacist` / `pharma123`**

Scope: CDSS allergy scan, Schedule X narcotic sign-off

💳 **POS Cashier: `cashier` / `cash123`**

Scope: 1-Click eRx import, FEFO batch, UPI QR, receipt

📦 **Inventory: `inventory` / `inv123`**

Scope: Multi-batch ledger, GRN inwarding, 2-8°C IoT

🛵 **Delivery Agent: `delivery` / `deliv123`**

Scope: Kanban dispatch, cold-bag check, OTP e-POD

RUBRIC: DEMONSTRATION OF IMPLEMENTED FUNCTIONALITY

Demonstration of Implemented Functionality

Phase 1: Clinical Pharmacist & CDSS Gate

Review incoming eRx queue. Trigger CDSS Allergy Interception (Penicillin warning blocks checkout). Execute Schedule X Narcotic double sign-off with Doctor DMC No + Pharmacist PIN.

Phase 2: POS Cashier Express Billing

1-Click import approved eRx into cart. Verify automatic FEFO nearest-expiry batch selection. Tender checkout: enter Cash (auto change calc) or Dynamic UPI QR. Print 80mm GST receipt.

Phase 3: Inventory FEFO & Cold-Chain IoT

Inspect batch ledger with countdown badges (<30d Red, <90d Yellow). Click 'Simulate Temp Spike' (9.4°C) on Chamber A fridge → observe immediate visual alert & log entry.

Phase 4: Fulfilment Fleet & OTP Handover

Advance order on Kanban dispatch board. Verify insulated cold-bag checklist. Enter customer 4-digit SMS OTP (1234) to confirm electronic Proof-of-Delivery (e-POD).

Phase 5: Admin Command Center & Z-Report Close

Monitor live terminal presence heartbeats. Open Daily Z-Report module: enter physical cash counted, verify variance (Balanced ₹0.00), sign and lock shift register.

REVIEW 1 & 2 SUMMARY, EMPIRICAL VERIFICATION & VIVA CLOSING

24 / 24 Automated Tests Passed (100% Reliability)

Review 1 Deliverables (Now Completed): Problem Statement, Quality Objectives, IEEE 830 SRS, 5 User Roles, Modular 6-Tier Architecture, 14 Tables, CRUD Matrix, Initial UI, Working Desktop & Web App, 24 Automated Tests.

Review 2 Horizon (Next Phase): Longitudinal FMEA occurrence tracking from error logs, automated real-time Pareto charts, Ishikawa incident registry, and multi-branch cloud database replication.

⭐ ONE SENTENCE FOR VIVA CLOSING

"The UI never writes SQL. Validation stops bad data. Audit records who did what. Expired stock cannot be sold. Logs become Pareto, FMEA, PDCA."

🙏 Sincere gratitude to the Computer Science & Management faculty at Amrapali University. The floor is open for viva questions. (Candidate: Piyush Bhatt)